

Generation Interconnection System Impact Study Report

For Queue Project Q18-1098

Covanta Indianapolis Energy 230 kV

December 22, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1098. The Interconnection Customer proposes a 175.4 MW solar photovoltaic facility interconnecting at the Covanta Indianapolis Energy 230 kV substation in the Ohio area. The request was received on November 04, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,967,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Northern States Flambeau / Franchise Tax 100.0 kV Ckt 1	Franchise Tax / Northern States Flambeau 100.0 kV #2	143.8	156	6.3
Waterford Power Llc / Myrtle-Date Substation 230.0 kV	Muskingum River / Tidd 500.0 kV	142.7	637	5.3
Northern States Flambeau / Franchise Tax 100.0 kV	Franchise Tax / Northern States Flambeau 100.0 kV	140.6	160	6.3
Filer 500-345 kV TX	Tidd / Lerdo Substation 500.0 kV	111.3	967	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Northern States Flambeau / Franchise Tax 100.0 kV Ckt 1	Rebuild line with higher-capacity conductor	\$10,252,000	100% (\$10,252,000)
Waterford Power Llc / Myrtle-Date Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$28,567,000	29% (\$8,378,000)
Northern States Flambeau / Franchise Tax 100.0 kV	Rebuild line with higher-capacity conductor	\$5,558,000	100% (\$5,558,000)
Filer 500/345kV Auto	Replace transformer with higher-rated unit	\$25,794,000	15% (\$3,779,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$27,967,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.